

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CCE) (Credit System) Degree Examinations**

January 2023

20CC503 – OPERATING SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer Five full questions choosing One full question from each Unit.***Unit – I**

- | | | Marks | BT* | CO* | PO* |
|-------|---|-------|-----|-----|-----|
| 1. a) | Define System call. Explain any two types of system calls. | 8 | L*2 | 1 | 1,2 |
| b) | Explain the transition from user to kernel mode with a neat diagram. | 6 | L2 | 1 | 1,2 |
| c) | Explain different states of a process with a neat diagram. | 6 | L2 | 1 | 1,2 |
| 2. a) | Discuss shared memory systems in accordance with inter-process communication. | 8 | L2 | 1 | 1,2 |
| b) | Explain the benefits of multithreaded programming. | 8 | L2 | 1 | 1,2 |
| c) | What is a process context switch? Explain in detail what happens when an OS does a process switch from a running process P0 to a ready to run process P1. | 4 | L3 | 1 | 1,2 |

Unit – II

3. a) Given below are the arrival and burst times of four processes P1, P2, P3 and P4. Draw the Gantt Chart using FCFS, Preemptive SJF, Calculate the average waiting time.

Processes	Burst time	Arrival Time
P1	8	0
P2	4	1
P3	9	2
P4	5	3

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|-------|--|----|----|---|-----|
| | | 10 | L3 | 2 | 1,2 |
| b) | Discuss dining philosopher problem solutions using monitors. | 10 | L3 | 2 | 1,2 |
| 4. a) | Consider the following snapshot of a system: | | | | |

	Allocation	Max	Available
	A B C D	A B C D	A B C D
P0	0 0 1 2	0 0 1 2	1 5 2 0
P1	1 0 0 0	1 7 5 0	
P2	1 3 5 4	2 3 5 6	
P3	0 6 3 2	0 6 5 2	
P4	0 0 1 4	0 6 5 6	

Answer the following questions using the banker's algorithm:

- What is the content of the matrix needed?
 - Is the system in a safe state?
 - If a request from process P1 arrives for (0, 4, 2, 0), can the request be granted immediately?
- | | | | | | |
|----|--|----|----|---|-----|
| | | 10 | L3 | 3 | 1,2 |
| b) | Explain about the methods used to prevent deadlocks. | 10 | L2 | 3 | 1,2 |

Unit – III

5. a) Given memory partitions of 100kB, 500kB, 200kB, 300kB, and 600kB (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212kB, 417kB, 112kB, and 426kB (in order)? Which algorithm makes the most efficient use of memory?
- | | | | | | |
|----|--|----|----|---|-----|
| | | 10 | L3 | 3 | 1,2 |
| b) | Explain FIFO Page Replacement, Discuss Belady's anomaly. | 10 | L3 | 4 | 1,2 |

P.T.O.

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|----|----|--|----|----|---|-----|
| 6. | a) | Write a note on the following with respect to virtual memory i) Thrashing ii) Local page replacement iii) Global page replacement. | 10 | L3 | 4 | 1,2 |
| | b) | Discuss paging. Explain the hardware support for paging with a paging hardware diagram. | 10 | L3 | 4 | 1,2 |

Unit – IV

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|----|----|---|----|----|---|-----|
| 7. | a) | Explain the six basic file operations. | 12 | L2 | 5 | 1,2 |
| | b) | Explain Virtual File System with a Schematic view diagram of a virtual file system. | 8 | L2 | 5 | 1,2 |
| 8. | a) | Write a note on i) Magnetic Disks ii) Magnetic Tapes. | 8 | L2 | 5 | 1,2 |
| | b) | Suppose a disk drive has 5000 cylinders numbered 0 to 4999. Drive is currently serving requests at cylinder 143, and previous request was at cylinder 125. The queue of pending requests in FIFO order is 86, 1470, 913, 1774, 948, 1509, 1022, 1750, 130. Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all pending requests for FCFS, SSTF, SCAN, CSCAN, LOOK disk scheduling algorithms. | 12 | L3 | 5 | 1,2 |

Unit – V

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|-----|----|---|----|----|---|-----|
| 9. | a) | Explain briefly various types of program threats and system threats. | 10 | L2 | 5 | 1,2 |
| | b) | Explain the access matrix model of protection and its implementation. | 10 | L2 | 5 | 1,2 |
| 10. | a) | What are the different authentication methods? Explain any two of them. | 10 | L2 | 5 | 1,2 |
| | b) | Discuss the use of password as a mechanism for protection. | 10 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
